

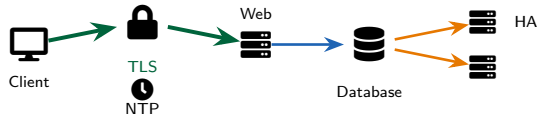
Module 7.0: Network Services and High Availability

Network+ Certification Course

Brendan Shea, PhD

RCTC

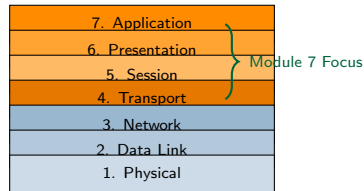
Introduction to Networking



Review: Building on Module 6

What We Learned (Module 6)

- **TCP/UDP:** Transport protocols and ports
- **DHCP:** DORA process for IP assignment
- **DNS:** Name resolution and record types
- **Ports:** Service identification (0–65535)



The Next Questions

How do we **secure** these services? How do we keep them **available**? How do we **synchronize time** across the network?

Module 7 Topics

TLS, NTP, HTTP(S), FTP, SMB, Email, VoIP, Disaster Recovery, High Availability

Module 7 Learning Objectives

1. Security & Time Sync

- Transport Layer Security (TLS)
- Network Time Protocol (NTP)
- Precision Time Protocol (PTP)



3. Email & Voice Services

- SMTP, IMAP, POP3
- VoIP protocols (SIP, RTP)
- QoS for voice traffic



2. Web, File & Database

- HTTP/HTTPS and web services
- FTP, SFTP, SMB file transfer
- NAS and database connectivity



4. DR & High Availability

- RTO, RPO, and DR sites
- Load balancing and clustering
- First Hop Redundancy (FHRP)



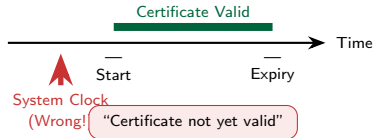
Section 7.1: Application Security and Time Synchronization

Why Application Security?

- Data in transit must be encrypted
- Authentication verifies identity
- Integrity ensures data isn't modified
- TLS provides all three!

Why Time Matters

- Certificate validation requires accurate time
- Log correlation across systems
- Kerberos authentication (5-min tolerance!)
- Scheduled tasks and backups



Key Insight

If your clock is wrong, **nothing works**—certificates fail, logs are useless, and authentication breaks!

Transport Layer Security (TLS) — Overview

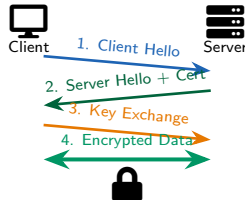
What is TLS?

TLS (Transport Layer Security) is a cryptographic protocol that provides:

- **Encryption:** Data privacy
- **Authentication:** Server (and optionally client) identity
- **Integrity:** Data hasn't been tampered with

TLS Versions

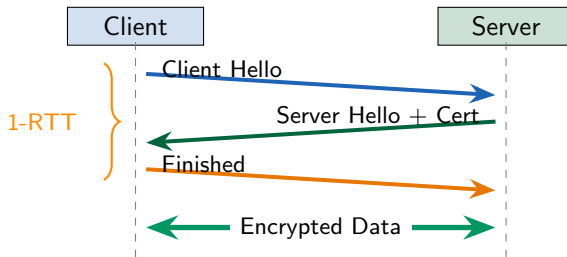
- SSL 2.0/3.0 — **Deprecated, insecure**
- TLS 1.0/1.1 — **Deprecated 2020**
- TLS 1.2 — Still widely used
- TLS 1.3 — Current standard (2018)



Common TLS Ports

HTTPS: 443	SMTPS: 465/587
IMAPS: 993	POP3S: 995

TLS 1.3 Handshake Process



TLS 1.3 Improvements

- 1-RTT handshake (faster than 1.2's 2-RTT)
- 0-RTT resumption for repeat connections
- Removed insecure cipher suites

Key Exchange

TLS 1.3 uses **Ephemeral Diffie-Hellman** exclusively—provides Perfect Forward Secrecy!

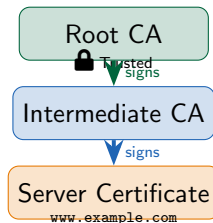
TLS Certificates and Trust

Certificate Chain

- **Root CA:** Pre-installed in browsers/OS
- **Intermediate CA:** Signed by Root
- **Server Certificate:** Signed by Intermediate

Certificate Fields

- **CN:** Common Name (domain)
- **SAN:** Subject Alternative Names
- **Validity:** Not Before / Not After dates
- **Public Key:** For encryption
- **Signature:** CA's digital signature



Self-Signed Certificates

Not trusted by browsers! Use for testing only.
Production sites need CA-signed certificates
(Let's Encrypt is free!).

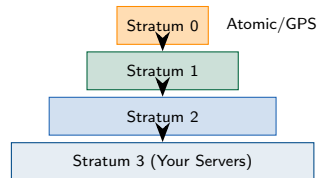
Network Time Protocol (NTP)

NTP Basics

- **Port:** UDP 123
- **Purpose:** Synchronize clocks across networks
- **Accuracy:** Typically within milliseconds
- **Polling:** Clients query servers periodically

Stratum Levels

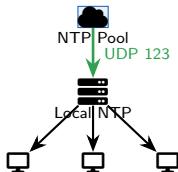
- **Stratum 0:** Atomic clocks, GPS (reference)
- **Stratum 1:** Directly connected to Stratum 0
- **Stratum 2:** Syncs from Stratum 1
- **Stratum 3+:** Each level adds latency



Public NTP Servers

pool.ntp.org
time.google.com
time.nist.gov
time.windows.com

NTP Architecture and Configuration



Why Local NTP Server?

Reduces external traffic, provides consistent time if internet fails.

Linux NTP Configuration

```
# /etc/chrony/chrony.conf
server pool.ntp.org iburst
server time.google.com iburst
```

Useful Commands

```
timedatectl status
chronyc sources -v
chronyc tracking
```

Kerberos Requirement

Kerberos requires clocks within **5 minutes**—NTP is essential for AD!

Precision Time Protocol (PTP)

PTP Overview

- **Standard:** IEEE 1588
- **Ports:** UDP 319 (event), 320 (general)
- **Accuracy:** Sub-microsecond to nanoseconds
- **Requirement:** Hardware timestamping support

PTP Use Cases

- Financial trading (timestamps matter!)
- Telecommunications (5G networks)
- Industrial automation
- Broadcast media synchronization

NTP vs PTP Comparison

Feature	NTP	PTP
Accuracy	ms	ns- μ s
Port	UDP 123	UDP 319/320
Hardware	Software	HW assist
Cost	Low	Higher
Complexity	Simple	Complex
Use Case	General	Precision

When to Use PTP

Only needed when **microsecond accuracy** is required. For most networks, NTP is sufficient and simpler!

Case Study: Dennis & The Paddy's Time Heist

"The Gang's POS System Fails"

Dennis notices Paddy's Pub POS system is suddenly rejecting all credit card transactions. The error message reads: **"Certificate not yet valid."**

Upon investigation, Dennis sees the system clock shows **January 15, 2019** instead of the current date. Mac admits he "fixed" the server by unplugging it during a storm last week and never bothered to check if it came back up properly.

```
System Time:      2019-01-15 14:32:00
Certificate Valid: 2023-06-01 to 2025-06-01
Error:            SSL_ERROR_FUTURE_CERT
```

Review Questions

- 1 Why are TLS certificates failing validation?
- 2 What protocol should be configured to prevent this?
- 3 How would you immediately fix this issue?

Case Study Solution: Dennis & The Paddy's Time Heist

Solution: "The Gang Learns About Time Sync"

- 1 System clock is in the **past** (2019), so certificates issued in 2023 appear "not yet valid"—the system thinks it's before the certificate start date!
- 2 Configure **NTP** to automatically synchronize with reliable time servers
- 3 **Immediate fix**: Manually set time or enable NTP sync

Quick Fix Commands

```
# Check current status
timedatectl status

# Enable NTP sync
sudo timedatectl set-ntp true

# Or use chronyd
sudo systemctl enable chronyd
sudo systemctl start chronyd
```

Key Lesson

"It's not just about the money, it's about sending a message... to the NTP server."

Certificates, Kerberos, logs, and scheduled tasks
all depend on accurate time!

HTTP — Hypertext Transfer Protocol

HTTP Basics

- **Port:** TCP 80 (unencrypted!)
- **Model:** Request-Response
- **Stateless:** Each request is independent
- Cookies/sessions add state

HTTP Methods

- **GET:** Retrieve resource
- **POST:** Submit data
- **PUT:** Update/replace
- **DELETE:** Remove resource

HTTP Status Codes

Code	Meaning
200	OK
301	Moved Permanently
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Server Error

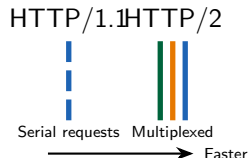
Security Warning

HTTP sends everything in **cleartext**. Never use for sensitive info!

HTTP/2 and HTTP/3

HTTP Version Comparison

Feature	1.1	2	3
Transport	TCP	TCP	QUIC
Multiplex	No	Yes	Yes
Header Comp	No	HPACK	QPACK
Server Push	No	Yes	Yes



HTTP/3 and QUIC

HTTP/3 uses **QUIC** (UDP-based) instead of TCP. Benefits: faster connection setup, better mobile performance, built-in encryption.

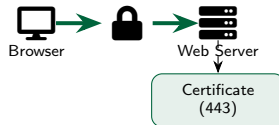
Key Takeaway

HTTP/2+ eliminates **head-of-line blocking**—multiple requests over single connection!

HTTPS — HTTP Secure

HTTPS = HTTP + TLS

- **Port:** TCP 443
- All HTTP traffic encrypted via TLS
- Server authenticated via certificate
- Data integrity protected



HSTS (HTTP Strict Transport Security)

Server header that forces browsers to **only use HTTPS**:

Strict-Transport-Security:
max-age=31536000; includeSubDomains

Verify Both!

Always check:

-  Padlock icon (encrypted)
- Domain name matches expected site

Phishing sites can have valid certificates!

FTP — File Transfer Protocol

FTP Basics

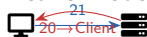
- **Control:** TCP port 21
- **Data:** TCP port 20 (active mode)
- Two separate connections
- **Cleartext credentials!**

FTP Modes

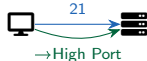
Active: Server connects TO client for data; **blocked by firewalls**

Passive (PASV): Client connects TO server; **firewall-friendly**

Active Mode



Passive Mode



Security Warning

FTP sends credentials in **cleartext**. Use SFTP or FTPS instead!

Secure File Transfer: SFTP, FTPS, and SCP

Secure File Transfer Comparison

Protocol	Port(s)	Encryption	Based On	Notes
SFTP	22	SSH	SSH subsystem	Single connection
FTPS	989/990	TLS	FTP + TLS	Explicit or Implicit
SCP	22	SSH	SSH/RCP	Simple copy only
FTP	20/21	None	—	Insecure

SFTP (Recommended)

- Uses SSH infrastructure (port 22)
- Single encrypted connection
- Full file operations
- Key-based auth supported

sftp user@server.com

FTPS Modes

Explicit: Start plain, upgrade via STARTTLS (port 21)

Implicit: TLS from start (port 990)

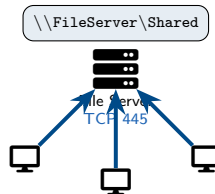
Firewall Note

FTPS still uses separate data connections. SFTP is simpler

SMB — Server Message Block

SMB Overview

- Windows file and print sharing
- **Port:** TCP 445 (direct)
- Legacy: TCP 137–139 (NetBIOS)
- UNC paths: `\\server\share`



SMB Versions

- **SMB1:** **Vulnerable! Disable it!**
- **SMB2:** Windows Vista/2008+
- **SMB2.1:** Windows 7/2008 R2
- **SMB3:** Windows 8/2012+ (encryption)
- **SMB3.1.1:** Windows 10/2016+ (pre-auth integrity)

CRITICAL: Disable SMB1!

SMB1 is vulnerable to **EternalBlue** exploit (WannaCry, NotPetya). Disable via:
`Disable-WindowsOptionalFeature`
`-FeatureName SMB1Protocol`

Network Attached Storage (NAS)

NAS Overview

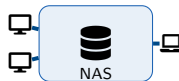
- **File-level** storage over network
- Dedicated storage appliance
- Protocols: SMB, NFS, AFP
- Easy setup and management
- Good for SMB (small-medium business)

Common NAS Protocols

- **SMB/CIFS**: Windows clients
- **NFS**: Linux/Unix clients (TCP 2049)
- **AFP**: Legacy Mac (TCP 548)
- **iSCSI**: Block-level (acts like SAN)

NAS vs SAN

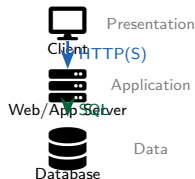
	NAS	SAN
Storage Level	File	Block
Protocol	SMB/NFS	FC/iSCSI
Network	Ethernet	Dedicated
Complexity	Low	High
Cost	\$	\$\$\$
Best For	File sharing	Databases



Database Services

Common Database Ports

Database	Port	Type
MySQL/MariaDB	3306	SQL
PostgreSQL	5432	SQL
MS SQL Server	1433	SQL
Oracle	1521	SQL
MongoDB	27017	NoSQL
Redis	6379	NoSQL/Cache



Security Best Practices

- Never expose DB directly to internet
- Use firewall rules to restrict access
- Encrypt connections (TLS)
- Use least-privilege accounts

Database Connectivity

- **ODBC**: Open Database Connectivity (Windows)
- **JDBC**: Java Database Connectivity
- Native drivers for each database

Case Study: Charlie's Rat Business Website

"Charlie's Rat Removal Goes Online"

Charlie has created a website for his new business "Charlie's Rat Removal & Bird Law Consulting" using plain HTTP on port 80.

Frank wants to add online payment processing to collect service fees. However, customers are complaining that their browsers display **"Not Secure"** warnings.

Charlie insists: "But it works fine on my computer! The rats don't care about security!"

Current URL: `http://charliesratremoval.com`

Port: `TCP 80`

Browser Warning: `'Not Secure'`

Payment Processor: `Refuses to integrate`

Review Questions

- 1 What is the security risk of using plain HTTP?
- 2 What needs to be implemented to fix this?
- 3 What port and protocol changes are needed?

Case Study Solution: Charlie's Rat Business Website

Solution: "Charlie Learns HTTPS"

- 1 HTTP transmits **everything in cleartext**—including payment info, passwords, and customer data. Anyone on the network can intercept it!
- 2 Implement **HTTPS** with a valid TLS certificate. Let's Encrypt provides free certificates!
- 3 Change from port **80 to 443**, configure TLS, and set up HTTP→HTTPS redirect.

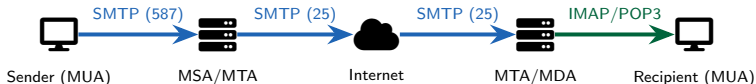
Apache HTTPS Redirect

```
<VirtualHost *:80>
  ServerName charliesratremoval.com
  Redirect permanent /
    https://charliesratremoval.com/
</VirtualHost>
```

Key Lesson

"The rats may not care, but PCI-DSS does!"
Payment Card Industry compliance **requires encryption**. No HTTPS = no payment processing!

Email Architecture Overview



Email Components

- **MUA:** Mail User Agent (client app)
- **MSA:** Mail Submission Agent
- **MTA:** Mail Transfer Agent (server-to-server)
- **MDA:** Mail Delivery Agent (to mailbox)

Store-and-Forward

Email uses **store-and-forward**: each server receives the message, stores it, then forwards to the next hop. Unlike real-time communication!

SMTP — Simple Mail Transfer Protocol

SMTP Basics

- **Port 25:** Server-to-server (MTA)
- **Port 587:** Client submission (MSA)
- **Port 465:** SMTPS (implicit TLS)
- **STARTTLS:** Upgrade to encrypted
- **Push protocol:** Sending only!

Email Authentication

- **SPF:** Authorized senders (DNS TXT)
- **DKIM:** Digital signature
- **DMARC:** Policy enforcement

Prevents spoofing and phishing!

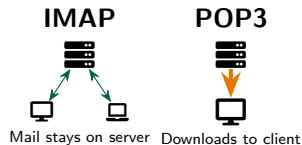
SMTP Session Example

```
S: 220 mail.example.com SMTP
C: HELO client.com
S: 250 Hello client.com
C: MAIL FROM:<sender@client.com>
S: 250 OK
C: RCPT TO:<user@example.com>
S: 250 OK
C: DATA
S: 354 Start mail input
C: Subject: Test
C:
C: Message body here.
C: .
S: 250 OK
C: QUIT
```


IMAP and POP3 — Receiving Email

IMAP vs POP3 Comparison

Feature	IMAP	POP3
Port (plain)	143	110
Port (TLS)	993	995
Mail Storage	Server	Downloads
Multi-device	Yes	Limited
Folders	Synced	Local only
Bandwidth	Higher	Lower
Offline	Sync folders	Full access
Server Space	Uses more	Uses less



When to Use

IMAP: Multiple devices, mobile users

POP3: Single device, archiving, limited server space

Voice and Video Services Overview

VoIP Overview

VoIP transmits voice as digital packets over IP networks instead of traditional phone lines (PSTN).

- Cost savings (no separate phone network)
- Unified Communications (UC)
- Advanced features (video, presence)
- Mobility and flexibility

QoS Requirements

Voice is **real-time**—sensitive to latency (<150ms), jitter (<30ms), and packet loss (<1%).

Traditional (PSTN)



VoIP



Quality Metrics

Metric	Target
One-way latency	<150ms
Jitter	<30ms
Packet loss	<1%
MOS Score	>4.0

VoIP Protocols

Protocol Reference

Protocol	Port	Purpose
SIP	5060/5061	Session setup
RTP	Dynamic	Media transport
H.323	1720	Legacy video
MGCP	2427/2727	Gateway control
SRTP	Dynamic	Secure RTP

SIP vs RTP

SIP: Signaling (call setup, teardown)—like dialing
RTP: Media (actual voice/video)—the conversation



RTP Ports

RTP uses **dynamic ports** (typically 16384–32767). Firewalls need to allow this range for VoIP!

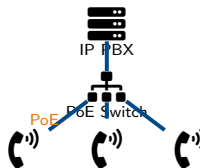
VoIP Phones and Infrastructure

VoIP Phone Types

- **Hard Phones:** Physical IP phones
- **Soft Phones:** Software on PC/mobile
- **ATA:** Analog Telephone Adapter

Infrastructure Requirements

- **PoE:** Power over Ethernet (802.3af/at)
- **Voice VLAN:** Separate from data
- **QoS:** DSCP EF (46) for voice
- **PBX/Call Manager:** Central control



Voice VLAN

Separate voice traffic onto its own VLAN:

- Isolates from data traffic
- Easier QoS policy application
- Security segmentation

Case Study: The Gang Gets a Phone System

“The Gang Upgrades to VoIP”

Paddy's Pub is upgrading to VoIP to save money. After installation, Dee complains calls sound “robotic” and choppy.

Investigation reveals:

- Mac downloading large files during business hours
- VoIP phones on same VLAN as all computers
- No QoS policies configured
- Switch doesn't support PoE

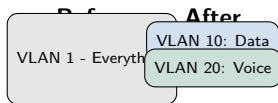
Review Questions

- 1 What is causing the voice quality issues?
- 2 What network configuration changes would help?
- 3 What QoS settings should be applied?

Case Study Solution: The Gang Gets a Phone System

Solution: “The Gang Learns QoS”

- 1 **Network congestion** causes jitter/latency—voice competes with downloads!
- 2 Create a **separate Voice VLAN** and implement **QoS policies**.
- 3 Configure **DSCP EF (46)** for voice; enable switch port trust.



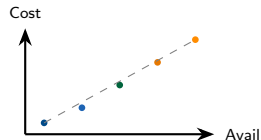
Key Lesson

Voice requires <150ms latency and minimal jitter. Without QoS, it competes equally with all other traffic!

Disaster Recovery and High Availability Concepts

Key Definitions

- **DR:** Getting back to normal *after* a failure
- **HA:** Preventing downtime *before* it happens
- **BC:** Keeping operations running during disruption



The “Nines” of Availability

Availability	Downtime/Year
99% (two 9s)	3.65 days
99.9% (three 9s)	8.76 hours
99.99% (four 9s)	52.6 min
99.999% (five 9s)	5.26 min

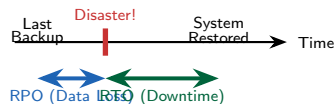
Cost vs Availability

Each additional “nine” roughly **10x the cost!** Match availability to business needs.

Disaster Recovery Metrics

Key DR Metrics

- **RTO** (Recovery Time Objective):
Maximum acceptable *downtime*
- **RPO** (Recovery Point Objective):
Maximum acceptable *data loss*
- **MTTR** (Mean Time To Repair): Average repair time
- **MTBF** (Mean Time Between Failures):
Reliability measure



Example Metrics by System

System	RPO	RTO
Email	24 hrs	4 hrs
E-commerce	1 hr	15 min
Trading	Real-time	Seconds

Disaster Recovery Sites

DR Site Types Comparison

Site Type	Equipment	Data	Recovery Time	Cost
Cold Site	None	None	Days–Weeks	\$
Warm Site	Partial	Recent backup	Hours–Days	\$\$
Hot Site	Full	Real-time sync	Minutes	\$\$\$
Cloud DR	On-demand	Replicated	Minutes–Hours	\$\$–\$\$\$

Cold Site

Empty facility with power/network. Must ship and install all equipment. Cheapest but slowest recovery.

Warm Site

Some equipment pre-installed. Need to restore data from backups. Balance of cost and speed.

Hot Site

Fully operational mirror. Real-time data replication. Immediate failover. Most expensive.

Fault Tolerance, Redundancy, and Load Balancing

Redundancy Technologies

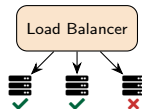
- **RAID:** Redundant disk arrays
- **NIC Teaming:** Multiple network adapters
- **Dual PSU:** Redundant power supplies
- **UPS:** Uninterruptible power
- **Multipathing:** Multiple storage paths

RAID Levels

RAID	Min	Tolerance
RAID 0	2	None (striping)
RAID 1	2	1 disk (mirror)
RAID 5	3	1 disk (parity)
RAID 10	4	1 per mirror

Load Balancing Methods

- **Round Robin:** Rotate through servers
- **Least Connections:** Send to least busy
- **IP Hash:** Same client → same server
- **Weighted:** Based on server capacity



Health Checks

Load balancers monitor server health and remove failed servers from the pool automatically!

First Hop Redundancy Protocols (FHRP)

The Problem

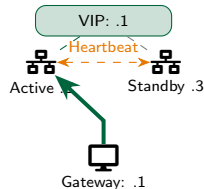
Clients use a single **default gateway**. If that router fails, clients lose connectivity!

FHRP Comparison

Protocol	Vendor	Load Bal
HSRP	Cisco	No
VRRP	Open	No
GLBP	Cisco	Yes

How It Works

Routers share a **Virtual IP (VIP)**. Clients point to VIP; active router responds. If active fails, standby takes over!



Transparent Failover

Clients **never change** their gateway—FHRP handles failover automatically!

Case Study: Frank's Foolproof Business Continuity Plan

"The Gang Needs Uptime"

Frank insists Paddy's Pub needs "five nines" availability for the new POS system after losing a full day of sales during the last power outage.

Current setup: Single server, no automated backups (Charlie "sometimes" copies files), one budget ISP, no UPS.

Frank's budget: "As little as possible!"

Solution: Realistic Improvements

- 1 **UPS** (\$150–300): Immediate protection, graceful shutdown
- 2 **Automated backups** (\$0–50/mo): Cloud backup for RPO
- 3 **Secondary ISP** (\$50–100/mo): If budget allows
- 4 **Realistic target:** 99.9% (three nines)—five nines costs five figures!

Module 7 Summary and Port Reference

Key Protocols by Section

7.1 Security/Time: TLS (443), NTP (123), PTP (319/320)

7.2 Web/File: HTTP (80), HTTPS (443), FTP (20/21), SFTP (22), SMB (445)

7.3 Email/Voice: SMTP (25/587), IMAP (143/993), POP3 (110/995), SIP (5060)

7.4 DR/HA: RTO, RPO, RAID, FHRP (HSRP/VRRP/GLBP)

Exam Reminders

- SFTP = SSH (port 22), not FTP!
- Disable SMB1 (security)
- Voice QoS: DSCP EF (46)
- IMAP = server storage, POP3 = download

Essential Port Reference

Service	Port	Proto
HTTP	80	TCP
HTTPS	443	TCP
FTP	20/21	TCP
SSH/SFTP	22	TCP
SMTP	25, 587	TCP
POP3/S	110, 995	TCP
IMAP/S	143, 993	TCP
NTP	123	UDP
SMB	445	TCP
MySQL	3306	TCP
SIP	5060/5061	Both